

KOLYVAN', Russian Federation

WMO: 296310

Lat: **55.322N**Lon: **82.739E**Elev: **147**StdP: **99.57**Time Zone: **7.00 (E07)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.5	-32.3	-38.1	0.1	-35.3	-35.1	0.1	-32.2	12.4	-10.4	10.5	-8.1	2.0	40	0.601

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	29.1	19.5	27.5	18.9	25.9	18.2	21.1	26.5	20.1	25.3	19.2	24.2	3.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.3	14.3	23.6	18.2	13.4	22.6	17.3	12.6	21.8	62.2	26.6	58.5	25.3	55.5	24.2	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.2	8.6	7.4	-36.8	32.1	4.2	1.8	-39.9	33.4	-42.4	34.4	-44.7	35.4	-47.8	36.7		
			-36.9	22.5	4.2	1.2	-39.9	23.4	-42.4	24.1	-44.8	24.8	-47.8	25.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.9	-17.8	-14.3	-6.1	3.7	11.6	17.4	19.1	16.6	10.3	2.9	-7.1	-14.4
	DBStd	14.22	8.49	7.32	6.21	5.70	5.63	4.45	3.23	3.66	4.63	5.17	7.52	8.31
	HDD10.0	3847	863	681	499	201	50	4	0	1	53	226	512	757
	HDD18.3	6127	1121	914	757	439	218	69	31	77	246	478	762	1016
	CDD10.0	891	0	0	0	12	99	226	281	206	61	7	0	0
	CDD18.3	130	0	0	0	0	9	41	54	23	3	0	0	0
	CDH23.3	1059	0	0	0	3	119	340	376	191	30	0	0	0
	CDH26.7	223	0	0	0	1	29	73	76	39	5	0	0	0
Wind	WSAvg	3.1	3.5	3.5	3.4	3.3	3.2	2.5	2.2	2.3	2.7	3.2	3.5	3.7
Precipitation	PrecAvg	470	24	19	21	27	34	57	70	58	41	44	39	36
	PrecMax	664	50	39	43	52	76	127	123	136	92	82	70	61
	PrecMin	330	5	2	3	6	1	18	9	10	15	13	22	16
	PrecStd	83	12	9	10	13	20	28	31	27	19	17	14	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.7	1.4	7.9	22.6	29.8	30.7	31.0	30.2	26.9	19.2	8.0	1.2
		MCWB	-0.3	0.0	3.8	12.5	17.1	19.7	21.5	20.0	17.2	11.6	6.3	0.3
	2%	DB	-2.1	-0.6	5.4	18.6	26.4	28.7	28.7	27.5	23.1	15.6	5.7	0.0
		MCWB	-3.0	-1.6	2.2	11.0	15.8	18.8	20.6	19.2	15.3	9.8	4.0	-0.5
	5%	DB	-3.8	-2.3	3.4	15.5	23.9	27.1	27.1	25.4	20.4	12.7	3.7	-1.6
		MCWB	-4.6	-3.3	1.1	9.1	14.4	18.3	19.9	18.2	14.0	8.5	2.5	-2.3
	10%	DB	-5.8	-4.1	1.9	12.5	21.4	25.2	25.5	23.4	17.8	10.1	1.7	-3.6
		MCWB	-6.6	-4.9	0.1	7.6	13.3	17.6	19.0	17.2	12.7	7.1	0.7	-4.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.1	0.4	4.5	13.4	18.6	21.3	23.1	22.0	17.9	12.2	6.8	0.6
		MCDB	0.8	1.1	7.6	20.3	28.0	28.1	28.8	26.5	24.4	17.5	7.8	0.8
	2%	WB	-2.9	-1.5	2.6	11.6	16.7	20.1	21.7	20.1	15.9	10.5	4.2	-0.5
		MCDB	-2.2	-0.7	4.6	17.9	24.9	26.5	26.9	25.3	21.8	14.5	5.2	0.0
	5%	WB	-4.6	-3.3	1.3	9.6	15.1	19.2	20.7	19.0	14.5	8.9	2.4	-2.3
		MCDB	-3.8	-2.4	3.0	14.9	22.4	25.2	25.5	24.0	19.6	12.4	3.6	-1.7
	10%	WB	-6.6	-4.9	0.2	7.6	13.7	18.3	19.7	17.9	13.1	7.3	0.8	-4.3
		MCDB	-5.9	-4.1	1.8	12.3	20.3	24.0	24.2	22.5	16.9	9.8	1.6	-3.6
Mean Daily Temperature Range	5% DB	MDBR	6.8	7.5	8.2	9.3	11.1	10.1	9.5	10.0	9.6	7.0	5.9	6.7
		MCDBR	7.4	7.1	8.4	13.8	14.4	12.4	11.4	12.7	13.3	11.7	5.6	6.6
		MCWBR	7.2	6.6	6.2	8.0	6.9	5.3	5.0	5.7	6.5	6.8	4.6	6.6
	5% WB	MCDBR	7.2	6.9	7.5	12.8	13.1	11.1	10.1	11.4	12.4	10.9	5.6	6.7
		MCWBR	7.1	6.6	6.0	8.0	7.1	5.4	5.1	5.8	6.6	6.7	4.8	6.8
Clear-Sky Solar Irradiance	taub	0.291	0.325	0.373	0.407	0.404	0.407	0.436	0.409	0.370	0.352	0.322	0.293	
	taud	2.006	1.967	1.916	2.081	2.220	2.290	2.290	2.208	2.270	2.327	2.242	2.055	1.954
	Ebn at Noon	583	707	763	806	837	839	802	801	783	688	536	467	
	Edh at Noon	70	106	143	141	130	123	131	116	95	81	67	59	
All-Sky Solar Radiation	RadAvg	0.91	1.93	3.42	4.62	5.29	5.96	5.34	4.47	3.00	1.51	0.73	0.57	
	RadStd	0.08	0.11	0.22	0.32	0.56	0.56	0.36	0.33	0.36	0.18	0.13	0.09	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	-0.61	N/A	N/A	N/A	N/A	N/A
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page